

Diffraction: The Single-Slit Pattern

PHYS 2203 — Introductory Physics III

September 16, 2026

Overview

We open by finishing last lecture's multi-slit interference, recapping the two-slit phasor sum and completing the general N -slit result. We then give the aperture finite width, and the structure that results is *diffraction*: the spreading of light as it passes through an aperture. Building on Huygens's principle, we model a single slit of width a as the $N \rightarrow \infty$ limit of the multi-slit problem and derive its intensity pattern.

1 Recap: Two-Slit Interference in the Complex Plane

Last lecture we found that two slits separated by d send fields to a screen point P that differ only by the phase

$$\delta = \frac{2\pi}{\lambda} d \sin \theta = k d \sin \theta, \quad (1)$$

picked up from the path difference $d \sin \theta$. The total field is

$$E(P, t) = E_0 [\sin(kr_1 - \omega t) + \sin(kr_1 - \omega t + \delta)]. \quad (2)$$

The two fields add in phase when the phase difference is a whole multiple of 2π , and cancel when it is an odd multiple of π . In terms of the path difference $d \sin \theta$,

$$\text{bright: } d \sin \theta = m\lambda, \quad \text{dark: } d \sin \theta = \left(m + \frac{1}{2}\right) \lambda, \quad m = 0, \pm 1, \pm 2, \dots \quad (3)$$

Evaluating the sum with the complex plane

Write the sum as the imaginary part of a sum of phasors:

$$E(P, t) = E_0 \operatorname{Im} \left\{ e^{i(kr_1 - \omega t)} (1 + e^{i\delta}) \right\}. \quad (4)$$

Adding the two phasors E_1 and E_2 (each of amplitude E_0 , with relative phase δ) head-to-tail forms an isosceles triangle whose base is the resultant.

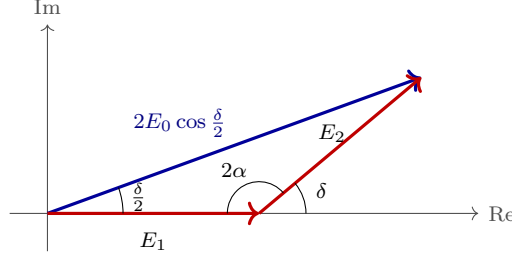


Figure 1: Phasor addition for $N = 2$, with E_1 drawn along the real axis. Each phasor has amplitude E_0 , and E_2 is turned from E_1 by the phase difference δ . The interior angle at the joint is 2α and the exterior turn is δ , so $2\alpha + \delta = \pi$ and $\alpha = \frac{\pi}{2} - \frac{\delta}{2}$. The resultant is the base of the isosceles triangle: it has length $2E_0 \cos(\delta/2)$ and makes angle $\delta/2$ with E_1 , so it equals $2E_0 \cos(\frac{\delta}{2}) e^{i\delta/2}$.

Resultant amplitude from the triangle. The interior angle at the joint plus the exterior turn δ make a straight line, so $2\alpha + \delta = \pi$, hence

$$\alpha = \frac{\pi}{2} - \frac{\delta}{2}. \quad (5)$$

Drop a perpendicular from the joint onto the resultant. Because the triangle is isosceles ($E_1 = E_2 = E_0$), this line bisects both the apex angle 2α and the base, splitting the triangle into two congruent right triangles, each with hypotenuse E_0 and apex half-angle α . The half-base is the side opposite α , so

$$\frac{1}{2} |E| = E_0 \sin \alpha = E_0 \sin \left(\frac{\pi}{2} - \frac{\delta}{2} \right) = E_0 \cos \frac{\delta}{2}, \quad (6)$$

using $\sin(\frac{\pi}{2} - x) = \cos x$. The resultant amplitude is therefore $|E| = 2E_0 \cos(\delta/2)$.

The field. From the triangle, the phasor sum $1 + e^{i\delta}$ is a vector of magnitude $2 \cos(\delta/2)$ (the resultant length, with the common E_0 factored out) pointing at angle $\delta/2$. Since a vector of magnitude M at angle φ is written $M e^{i\varphi}$,

$$1 + e^{i\delta} = \underbrace{2 \cos \frac{\delta}{2}}_{\text{magnitude}} \underbrace{e^{i\delta/2}}_{\text{rotated by } \delta/2}. \quad (7)$$

Substituting this and evaluating the imaginary part,

$$\begin{aligned} E(P, t) &= E_0 \operatorname{Im} \left\{ e^{i(kr_1 - \omega t)} (1 + e^{i\delta}) \right\} \\ &= \operatorname{Im} \left\{ 2E_0 \cos \frac{\delta}{2} e^{i(kr_1 - \omega t + \delta/2)} \right\} \\ &= 2E_0 \cos \frac{\delta}{2} \sin \left(kr_1 - \omega t + \frac{\delta}{2} \right). \end{aligned} \quad (8)$$

The intensity is the time average of E^2 , and since $\langle \sin^2 \rangle = \frac{1}{2}$ is a constant,

$$\text{Intensity} \sim \langle E^2(P, t) \rangle \propto \left(2E_0 \cos \frac{\delta}{2} \right)^2, \quad (9)$$

which is maximized when $\frac{\delta}{2} = m\pi$, i.e.

$$\frac{1}{2} \frac{2\pi}{\lambda} d \sin \theta = m\pi \implies d \sin \theta = m\lambda \quad \checkmark \quad (10)$$

2 Multiple Slits: The Geometric Series

For N slits the field is again the imaginary part of a phasor sum, now with N terms:

$$E(P, t) = E_0 \operatorname{Im} \left\{ e^{i(kr_1 - \omega t)} (1 + e^{i\delta} + e^{i2\delta} + \dots + e^{i(N-1)\delta}) \right\}, \quad \delta = \frac{2\pi}{\lambda} d \sin \theta. \quad (11)$$

For two slits we added the phasors as vectors and read the resultant off the triangle — the “use geometry” route. For general N that construction becomes unwieldy, but the bracket is simply a **geometric series** — the “use algebra” route.

Sum the series. Let $r = e^{i\delta}$. Each term is then a power of r , since $e^{in\delta} = r^n$, so the sum is a geometric series:

$$\sum_{n=0}^{N-1} e^{in\delta} = \sum_{n=0}^{N-1} r^n = \frac{r^N - 1}{r - 1} = \frac{e^{iN\delta} - 1}{e^{i\delta} - 1}. \quad (12)$$

Read off the magnitude. Factor a half-angle out of $e^{ix} - 1$:

$$e^{ix} - 1 = e^{ix/2} (e^{ix/2} - e^{-ix/2}). \quad (13)$$

The bracket equals $2i \sin(x/2)$. To see this, subtract the two Euler relations

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad e^{-i\theta} = \cos \theta - i \sin \theta \quad (14)$$

term by term: the cosines cancel and the $i \sin$ terms add, so $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$ (here $\theta = x/2$). Hence

$$e^{ix} - 1 = e^{ix/2} (2i \sin \frac{x}{2}). \quad (15)$$

Applying this to the numerator ($x = N\delta$) and the denominator ($x = \delta$),

$$\frac{e^{iN\delta} - 1}{e^{i\delta} - 1} = \frac{e^{iN\delta/2} (2i) \sin \frac{N\delta}{2}}{e^{i\delta/2} (2i) \sin \frac{\delta}{2}} = e^{i(N-1)\delta/2} \frac{\sin \frac{N\delta}{2}}{\sin \frac{\delta}{2}}. \quad (16)$$

The field is therefore

$$E(P, t) = E_0 \frac{\sin \frac{N\delta}{2}}{\sin \frac{\delta}{2}} \sin \left(kr_1 - \omega t + \frac{(N-1)\delta}{2} \right), \quad (17)$$

and the intensity is

$$\text{Intensity} \sim E^2 = E_0^2 \frac{\sin^2 \left(\frac{N\delta}{2} \right)}{\sin^2 \left(\frac{\delta}{2} \right)}. \quad (18)$$

Check, $N = 2$. $\frac{\sin \delta}{\sin(\delta/2)} = \frac{2 \sin \frac{\delta}{2} \cos \frac{\delta}{2}}{\sin \frac{\delta}{2}} = 2 \cos \frac{\delta}{2}$, recovering the two-slit triangle result — the algebra and the geometry agree.

For $N = 6$, the ratio of these two functions produces the intensity pattern below.

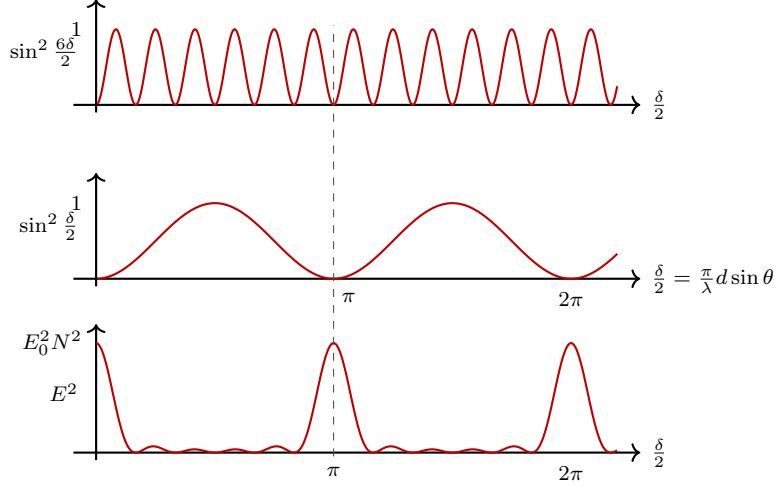


Figure 2: $N = 6$ intensity construction. The numerator $\sin^2(N\delta/2)$ oscillates rapidly; the denominator $\sin^2(\delta/2)$ vanishes at $\delta/2 = m\pi$. Their ratio E^2 produces tall principal maxima at $\delta/2 = m\pi$ (height $E_0^2 N^2$) separated by weak subsidiary peaks.

3 The 0/0 Limit at the Principal Maxima

When $\frac{\delta}{2} = (\text{integer})\pi$, both numerator and denominator vanish and the formula gives $\frac{0}{0}$. There are two ways to resolve this:

1. Apply L'Hôpital's rule (twice).
2. Evaluate E^2 for $\frac{\delta}{2} = (\text{integer})\pi + \Delta$ where Δ is very small.

Method 2. Using $\sin^2[(\text{integer})\pi + \Delta] = \sin^2[\Delta] \approx \Delta^2$, and similarly $\sin^2(\frac{N\delta}{2}) \approx (N\Delta)^2$ when $N\Delta$ is very small,

$$E^2 = E_0^2 \frac{\sin^2(\frac{N\delta}{2})}{\sin^2(\frac{\delta}{2})} \approx E_0^2 \frac{\sin^2(N\Delta)}{\sin^2(\Delta)} \approx E_0^2 \frac{(N\Delta)^2}{\Delta^2} = \boxed{E_0^2 N^2} \quad (19)$$

since $\sin \theta \approx \theta$ when θ is small. The peak intensity at each principal maximum therefore grows as N^2 .

Consequence. Larger $N \rightarrow$ the principal peaks are more intense and narrower.

This is the idea behind the **diffraction grating**: use N slits to spatially separate non-monochromatic light. The principal maxima occur at $d \sin \theta = m\lambda$, so different wavelengths emerge at different angles, allowing the component colors to be cleanly resolved.

4 Huygens's Principle and Diffraction

Diffraction is the spreading of light as it passes through an aperture.

Huygens's principle (1600s). This is a geometrical construction: *each point on a wave front acts as a source of a wavelet that propagates outward. The wave front at some later time is the envelope of the wavelets.* We have already drawn this picture: the circular wavelets radiating from the narrow slits in last lecture's two-slit figure are exactly Huygens wavelets from point-like apertures.

Where does this come from? It is a direct consequence of the wave equation, but the derivation is too advanced for this class.

5 Single-Slit Diffraction

Consider a plane wave incident on a slit of width a . Light passing through the slit spreads out and forms an intensity pattern on a distant screen, with a bright central maximum flanked by weaker side peaks.

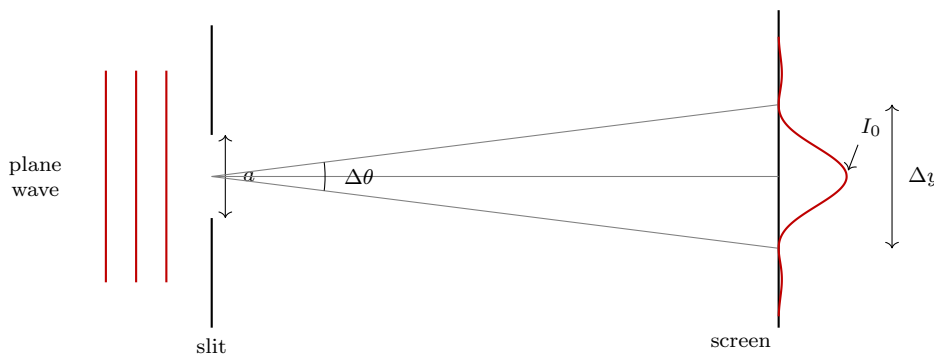


Figure 3: Single-slit diffraction. A plane wave incident on a slit of width a produces a diffraction pattern on a distant screen, with a bright central maximum of peak intensity I_0 , full width Δy between the first minima, and angular spread $\Delta\theta$.

Treating the slit as N sources

Model the slit as N point sources with a spacing $d = a/N$. Eventually we take $d \rightarrow 0$ and $N \rightarrow \infty$ so that the discrete sources become a continuous aperture.

For N equally spaced coherent sources, the intensity at the point P is

$$I \sim E_0^2 \frac{\sin^2\left(\frac{N\delta}{2}\right)}{\sin^2\left(\frac{\delta}{2}\right)}, \quad \text{where} \quad \delta = \frac{2\pi}{\lambda} d \sin \theta \quad (20)$$

is the phase difference between adjacent sources.

The combinations that appear in the limit are

$$\frac{N\delta}{2} = \frac{N}{2} \frac{2\pi}{\lambda} d \sin \theta = \frac{\pi a}{\lambda} \sin \theta, \quad (21)$$

$$\frac{\delta}{2} = \frac{\pi d}{\lambda} \sin \theta \quad (\text{very small}), \quad (22)$$

where in the first line we used $Nd = a$. Name the numerator's argument

$$\alpha \equiv \frac{N\delta}{2} = \frac{\pi a}{\lambda} \sin \theta, \quad \text{so that} \quad \frac{\delta}{2} = \frac{\alpha}{N}. \quad (23)$$

Now handle the two $\sin^2$ factors separately.

Numerator. It is already exact — no approximation:

$$\sin^2\left(\frac{N\delta}{2}\right) = \sin^2 \alpha. \quad (24)$$

Denominator. Because $d = a/N$, the angle $\frac{\delta}{2} = \frac{\alpha}{N}$ becomes tiny as $N \rightarrow \infty$, so $\sin \frac{\delta}{2} \approx \frac{\delta}{2}$ and

$$\sin^2\left(\frac{\delta}{2}\right) \approx \left(\frac{\delta}{2}\right)^2 = \left(\frac{\alpha}{N}\right)^2 = \frac{\alpha^2}{N^2}. \quad (25)$$

Ratio. Dividing the numerator by the denominator,

$$I \sim E_0^2 \frac{\sin^2 \alpha}{\alpha^2/N^2} = N^2 E_0^2 \frac{\sin^2 \alpha}{\alpha^2}. \quad (26)$$

Each of the N sources carries $1/N$ of the slit's light, so its amplitude scales as $E_0 = E_{\text{tot}}/N$. Then $N^2 E_0^2 = (NE_0)^2 = E_{\text{tot}}^2$ is finite and independent of both N and θ — the vanishing E_0 exactly cancels the growing N^2 . This constant sets only the overall brightness, so the shape of the diffraction pattern is

$$\boxed{I \sim \frac{\sin^2 \alpha}{\alpha^2}}, \quad \text{where} \quad \alpha = \frac{\pi a}{\lambda} \sin \theta. \quad (27)$$

6 The Single-Slit Intensity Pattern

The intensity has the limiting values

$$I \sim \frac{\sin^2 \alpha}{\alpha^2} = \begin{cases} 1 & \text{when } \alpha = 0, \\ 0 & \text{when } \alpha = \pm\pi, \pm2\pi, \dots \end{cases} \quad (28)$$

The central maximum (at $\alpha = 0$) is bright, and the intensity vanishes at the integer multiples of π , between which lie progressively weaker secondary maxima.

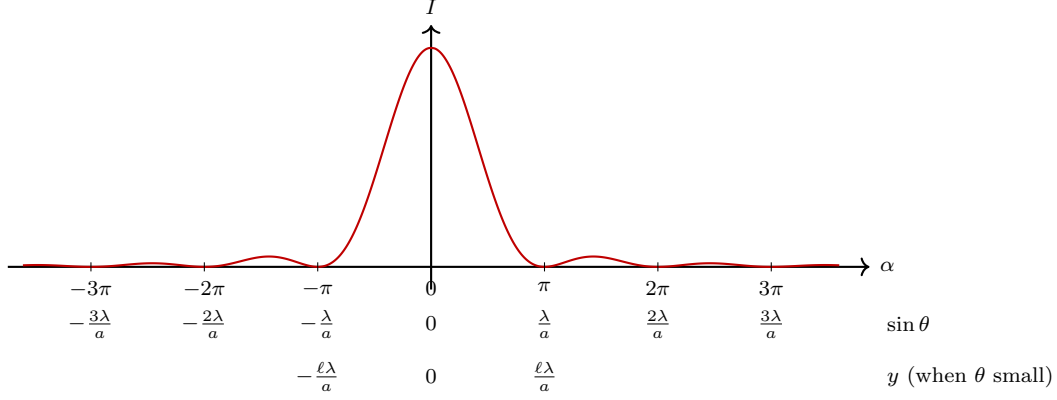


Figure 4: Single-slit intensity $I \sim \sin^2 \alpha / \alpha^2$ versus α . The lower axes give the corresponding values of $\sin \theta = \alpha \lambda / (\pi a)$ and, for small angles, the screen position $y \approx \ell \tan \theta$. Minima occur at $\sin \theta = m \lambda / a$ for nonzero integer m .

Angular variables and the width of the central peak

We derived the pattern in terms of α , but on a screen we measure either the angle θ from the slit or the position y on the screen. The three are linked by

$$\alpha = \frac{\pi a}{\lambda} \sin \theta, \quad y = \ell \tan \theta \approx \ell \theta \quad (\theta \text{ small}), \quad (29)$$

where ℓ is the slit-to-screen distance. Any feature of the pattern can therefore be quoted in α , in $\sin \theta$, or in y — these are the three axes of the figure above.

Locating the first minimum. The first dark fringe is at $\alpha = \pi$. Converting to an angle with $\alpha = \frac{\pi a}{\lambda} \sin \theta$,

$$\alpha = \pi \implies \frac{\pi a}{\lambda} \sin \theta_1 = \pi \implies \sin \theta_1 = \frac{\lambda}{a}. \quad (30)$$

For small angles $\theta_1 \approx \lambda/a$, and its position on the screen is $y_1 = \ell \theta_1 \approx \ell \lambda / a$.

Width of the central peak. The bright central maximum runs from the first minimum on one side to the first minimum on the other, so its full width is twice y_1 :

$$\Delta y = 2y_1 \approx \frac{2\ell\lambda}{a}. \quad (31)$$

As a full *angular* width, $\Delta\theta = 2\theta_1$:

$$\Delta\theta = 2 \sin^{-1} \left(\frac{\lambda}{a} \right) \approx \frac{2\lambda}{a} \quad (\theta \text{ small}), \quad (32)$$

consistent with $\Delta y = \ell \Delta\theta$.

What this tells us. Both widths grow as the slit *narrows* ($\Delta y, \Delta\theta \propto 1/a$): a smaller a gives a *wider* central peak. That is the counterintuitive signature of diffraction — squeezing the beam makes it spread more. It also shows *when* diffraction matters:

- If $a \gg \lambda$, then λ/a is tiny, so $\Delta\theta \approx 2\lambda/a \rightarrow 0$: the light goes essentially straight through and casts a sharp geometric shadow. Diffraction is negligible.
- If $a \sim \lambda$, then $\lambda/a \sim 1$ and $\Delta\theta$ is large: the light fans out over a wide angle. Diffraction dominates.

So diffraction is most evident when $a \sim \lambda$ or smaller.